

1. Administrative & Study Identification

Full Study Title: Substudy 01I: A Study of Investigational Agents in Participants With Previously Treated Stage IV Squamous Non-small Cell Lung Cancer (NSCLC) (MK-3475-01I/KEYMAKER-U01I)

Official Title: KEYMAKER-U01 Substudy 01I: A Phase 2, Randomized, Umbrella Study With Rolling Arms of Investigational Agents in Participants With Previously Treated Stage IV Squamous Non-small Cell Lung Cancer (NSCLC) **Short Title/Acronym:** KEYMAKER-U01

Substudy 01I Protocol Number: MK-3475-01I / 3475-01I **NCT Number:** NCT06780098 **Trial Status:** RECRUITING **Sponsor/Company Name:** Merck (Merck Sharp & Dohme LLC) **Investigational Product:** Ifinatamab deruxtecan (I-DXd), Raludotatug deruxtecan (R-DXd), and Docetaxel **Phase of Development:** Phase 2 **Medical Monitor:** Merck Sharp & Dohme LLC Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the cancer response to the study treatments (Antibody Drug Conjugates: I-DXd and R-DXd) compared to chemotherapy, the safety of the study treatments, and whether participants tolerate them in individuals with previously treated Stage IV Squamous NSCLC. **Secondary Objectives:** To evaluate the Duration of Response (DOR), Progression-Free Survival (PFS), and Overall Survival (OS).

2.2 Background & Rationale Standard treatment for metastatic squamous NSCLC (cancer that starts in flat squamous cells lining the airways) is immunotherapy with or without chemotherapy. However, standard treatments may fail or stop working. Researchers are investigating whether targeted study treatments, specifically antibody drug conjugates (ADCs) which attach to proteins on cancer cells and deliver treatment to destroy them, can effectively treat metastatic squamous NSCLC that did not respond to standard regimens.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Docetaxel) **Blinding:** Single-blind

Randomization: Randomized **Master Protocol:** This study is one of the substudies conducted under the pembrolizumab umbrella master screening protocol MK-3475-U01 (KEYMAKER-U01) - NCT04165798.

3.2 Planned Enrollment & Duration Target Enrollment (\$N\$): 144 participants **Number of Sites:** Multiple global sites **Estimated**

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Histologically or cytologically confirmed diagnosis of Stage IV squamous non-small cell lung cancer (NSCLC). 2. Has documented disease progression per Response Evaluation Criteria In Solid Tumors 1.1 (RECIST 1.1), assessed by the investigator after receiving an anti-programmed cell death protein 1 (anti-PD-1) or anti-PD-L1 treatment and platinum-based chemotherapy for Stage IV disease. 3. Has measurable disease per RECIST 1.1. 4. Human immunodeficiency virus (HIV)-infected participants must have well-controlled HIV on antiretroviral therapy (ART). 5. Participants who are Hepatitis B surface antigen (HBsAg) positive are eligible if they have received Hepatitis B virus (HBV) antiviral therapy for at least 4 weeks, and have undetectable HBV viral load. 6. Participants with history of Hepatitis C virus (HCV) infection are eligible if HCV viral load is undetectable.

4.2 Exclusion Criteria 1. Diagnosis of small cell lung cancer or, for mixed tumors, the presence of small cell elements. 2. Uncontrolled or significant cardiovascular disorder. 3. History of (noninfectious) pneumonitis or Interstitial Lung Disease (ILD) requiring steroids, or current suspected/active pneumonitis/ILD. 4. Clinically severe pulmonary compromise resulting from intercurrent pulmonary illnesses including, but not limited to, any underlying pulmonary disorder (e.g., pulmonary emboli within 3 months, severe asthma, severe chronic obstructive pulmonary disease, restrictive lung disease, pleural effusion) or any autoimmune, connective tissue, or inflammatory disorders with pulmonary involvement, or prior pneumonectomy. 5. Participants who have adverse events (AEs) (other than alopecia) due to previous anticancer therapies must have recovered to $\leq$ Grade 1 or baseline. 6. Evidence of ongoing uncontrolled systemic bacterial, fungal, or viral infection (including HIV infection). 7. Has clinically significant corneal disease. 8. Known additional malignancy that is progressing or has required active treatment within the past 3 years. 9. Has known untreated central nervous system (CNS) metastases and/or carcinomatous meningitis. 10. Evidence of any leptomeningeal disease. 11. Active autoimmune disease requiring systemic treatment in the past 2 years (replacement therapy is allowed). 12. Active infection requiring systemic therapy. 13. HIV-infected participants with a history of Kaposi's sarcoma and/or Multicentric Castleman's Disease. 14. Active inflammatory bowel disease requiring immunosuppressive medication or previous clear history of inflammatory bowel disease (e.g., Crohn's disease, ulcerative colitis). 15. Known history of, or active, neurologic paraneoplastic syndrome. 16. History of allogeneic tissue/solid organ transplant. 17. Has not

adequately recovered from major surgery or have ongoing surgical complications.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Ifinatumab deruxtecan (I-DXd) at 8 mg/kg or 12 mg/kg; Raludotatug deruxtecan (R-DXd); or Docetaxel (Trade names: Docefrez, Taxotere). Administered according to pre-specified dosing cycles. **Route of Administration:** Intravenous (IV) Infusion. **Duration of Treatment:** Until documented disease progression, unacceptable toxicity, investigator decision, or study completion.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care medications to manage side effects. **Prohibited:** Concurrent antineoplastic therapy, active systemic therapies for ongoing infections, or live-attenuated vaccines.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Up to Day -28 to 0	Informed Consent, Medical History, Baseline RECIST 1.1 Scans, Labs
Baseline	Day 0	Randomization, First Dose
Interim	Per Treatment Cycle	Safety Labs, Efficacy Assessment, IV Infusion, AE Monitoring
End of Study	30 days post last dose	Final Vital Signs, Exit Interview, Safety Follow-Up, Survival Follow-Up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints 1. **Objective Response Rate (ORR):** Defined as the percentage of participants achieving a Complete Response (CR) or Partial Response (PR) assessed per RECIST 1.1 by Blinded Independent Central Review (BICR) and by the investigator. 2. **Adverse Events (AEs):** Number of participants who experience one or more adverse events (AEs). 3. **Discontinuation due to AE:** Number of participants who discontinue study intervention due to an AE.

7.2 Secondary & Exploratory Endpoints 1. **Progression-Free Survival (PFS):** Assessed by BICR. 2. **Duration of Response (DOR):** Assessed by BICR. 3. **Overall Survival (OS).**

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection of AEs and Treatment-Related Adverse Events (TRAEs), tracked during

treatment and up to 30 days after the last dose. **Serious Adverse Events (SAEs):** Life-threatening events or serious toxicities reported within 24 hours of investigator awareness. **Data Monitoring Committee (DMC):** External/Internal committees will evaluate periodic safety profiles and efficacy signals during rolling arms.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) set for principal efficacy evaluations (ORR, PFS, OS). Safety Set for all participants receiving at least one dose of the study drug.

Sample Size Justification: Target enrollment of $N=144$ is statistically powered to detect meaningful shifts in ORR compared to the active comparator docetaxel. **Handling of Missing Data:** Standard survival analysis censoring rules applied to time-to-event endpoints (PFS, OS, DOR).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Scheduled onsite and remote monitoring to assure protocol adherence. **Audit Trail:** Secure electronic Case Report Form (eCRF) locking and independent auditing of imaging and pathology (BICR).

11. Study Identifiers & Governance

Primary ID: MK-3475-01I **Registry Number:** NCT06780098 **Posting ID:** 548383055385934621 **Sponsor/Lead Organization:** Merck **Study Title:** KEYMAKER-U01 Substudy 01I **Official Regulatory Title:** KEYMAKER-U01 Substudy 01I: A Phase 2, Randomized, Umbrella Study With Rolling Arms of Investigational Agents in Participants With Previously Treated Stage IV Squamous Non-small Cell Lung Cancer (NSCLC)

12. Clinical Framework (Oncology Specific)

Disease Area: Lung Neoplasms **Primary Condition:** Stage IV Squamous Non-small Cell Lung Cancer (NSCLC) **Preferred UMLS Name:** Squamous Cell Carcinoma of Lung **Diagnosis Threshold:** Confirmed disease progression following prior anti-PD-1/PD-L1 and platinum-based chemotherapy **Medical Methodology:** Antibody Drug Conjugate (ADC) targeted therapy vs. standard taxane chemotherapy **Optimization Target:** Objective Response Rate (ORR) and maximum tumor size reduction

13. Study Procedures & Methodology

Management Pathway: Standard clinical pathway for advanced/metastatic squamous NSCLC progressing on 1L therapy. **Education Protocol (HCP):** ADC administration, recognizing early symptoms of treatment-related pneumonitis/ILD, and infusion reaction management. **Education Protocol (Patient):** Informed consent, treatment timeline, and toxicity/symptom self-reporting. **Quality Audit Mechanism:** Blinded Independent Central Review (BICR) for RECIST 1.1 tumor evaluations.

14. Observation & Follow-Up Schedule

Total Duration: Up to approx. 7 years (Primary completion estimated March 2032) **Onsite Visit Cadence:** Regular intervals aligning with IV infusion cycles **Remote Monitoring Frequency:** As required between active infusion visits **Checklist Review Interval:** Routine imaging/scans conducted every 6-9 weeks per standard RECIST 1.1 monitoring

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				